6.808 Mobile and Sensor Computing

aka IoT Systems

Lecture #6
Mesh Networks & Multi-Hop Routing

- Pset 1 due March 5
- Laptops/iPads are fine if you are taking notes for the class (but not for other work)

Today in IoT

Snowstorm live weather forecast updates as Boston area snow totals, power outages, blizzard conditions rise

By Mike Toole, Matt Schooley, Neal Riley

Updated on: February 23, 2026 / 3:33 PM EST / CBS Boston

Add CBS News on Google

Blizzard of 2026 could be historic snowstorm in Massachusetts

- Snow totals topped more than two feet in southeastern Massachusetts. A travel ban was issued there.
- Power outages are a major problem. Wind gusts topped 80 miles per hour in Massachusetts.

How are concepts from this class used?

1. Weather prediction: IoT sensors (balloons/radiosondes), satellites (clouds), weather stations (doppler radars), commercial aircraft
2. Emergency response: power grid fault prediction + rerouting, flood warning system

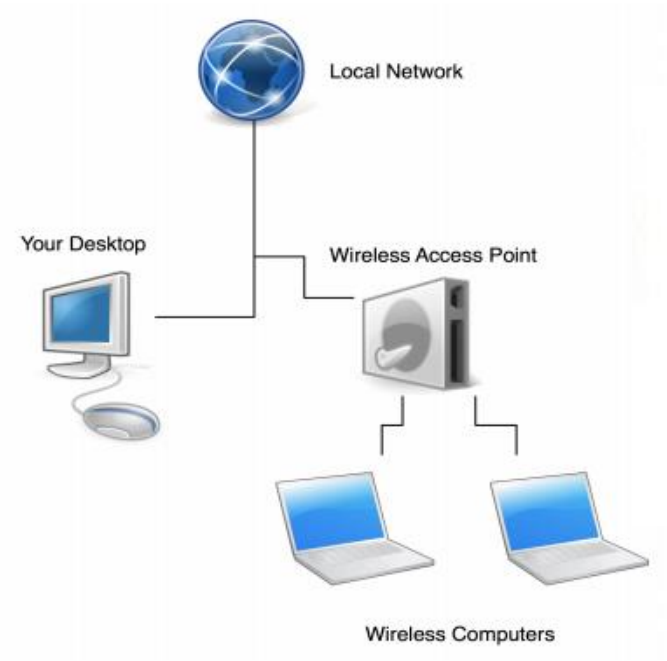
How is snow total measured?

- ultrasonic/ laser-depth
- Weighing gauges

Wireless Network Architectures

There are 3 kinds of wireless network architectures

Access Network

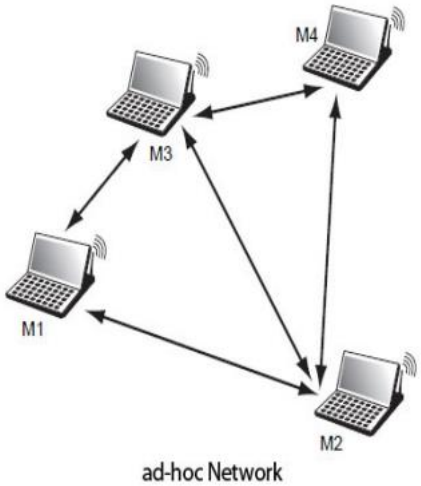


e.g., WiFi, cellular

Device-to-device

e.g., Bluetooth

Ad Hoc Network



e.g., leverage P2P to reach internet (crises)

One-hop

Multi-hop

RoofNet

MIT Technology Review

Networking From the Rooftop

MIT researchers are developing new routing strategies for a wireless network that hops data in the roofs of the city.

by **Erico Guizzo** Aug 29, 2003

A few weeks ago, MIT graduate student Shan Sinha canceled his broadband Internet service. Now his Net connection comes through the chimney. From

7 YEARS AFTER ROOFNET, MIT AND CSAIL CHOOSE MERAKI FOR WIRELESS LAN

February 17, 2010
Posted by: @merakisimon
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TC

Cisco Acquires Enterprise Wi-Fi Startup Meraki For $1.2 Billion In Cash

Josh Constine @joshconstine / 6:36 pm EST • November 18, 2012 Comment

The Tech Established 1881 ONLINE EDITION

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Volume 125 >> Issue 65 : Wednesday, February 1, 2006 PDF of This Issue

MIT and City Collaborate To Provide Free ess

Thibault
EWS EDITOR

n with MIT researchers may provide Cambridge with a free, city-wide, wireless internet service as early er. The project will rely on a mesh networking technology that allows individual computers to become oints, projecting the reach of the network beyond its original antennas.

l of the project is to provide internet access to Cantabrigians who live in public housing, said Cambridge tion Officer Mary P. Hart, though the resulting infrastructure will have a far wider benefit for city

chow '68, vice president for Information Services and Technology, said he expects the maximum speed to be 54 megabits per second. The speed users experience will decline as more people access the

although the level of internet service will not be known until the antennas are tested, users should be a browser and send e-mail, though they might not be able to send large pictures or view streaming

Single Path Routing

Represent the wireless network as a graph

- Two nodes have an edge if they can communicate (i.e., are within radio range)
- Each edge is labeled with a weight (where a smaller weight indicates a preferred edge)

Run shortest path algorithm on the graph (e.g., Dijkstra)

- Produce the minimum weight path between every pair of nodes

How do you pick the edge weights?

- i.e., what metric should shortest path minimize?

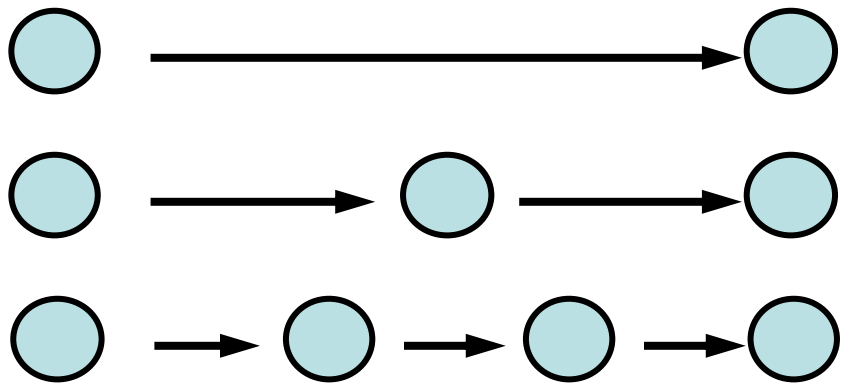
Focus of this lecture

Approach 1:

Assign all edges the same weight → Minimize number of hops

Reasoning:

- Links in route share radio spectrum
- Extra hops reduce throughput



Throughput = ?

Approach 1:

Assign all edges the same weight → Minimize number of hops

Reasoning:

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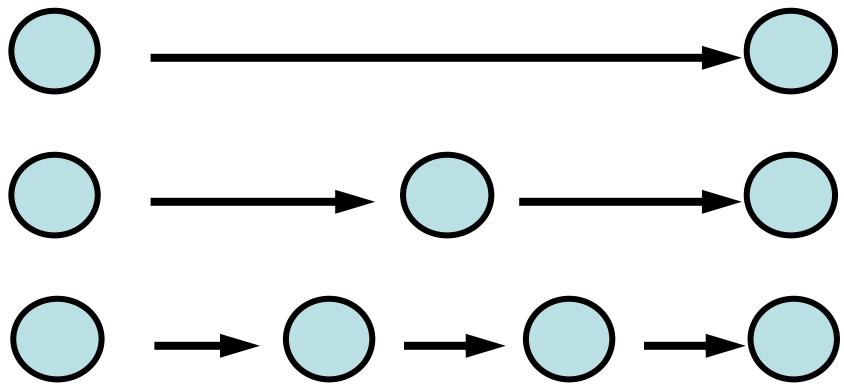


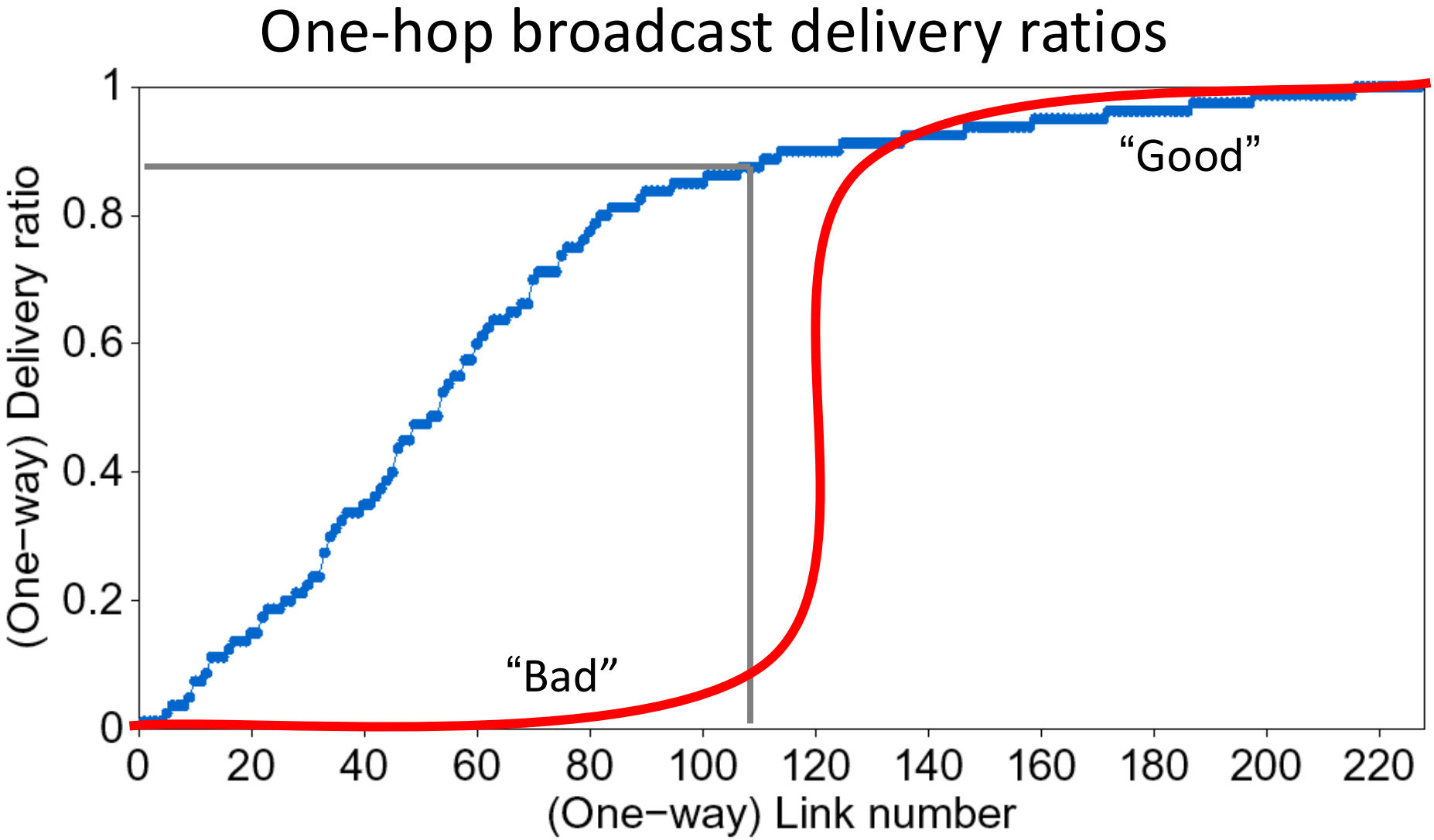
Throughput = 1

Throughput = 1/2

Throughput = 1/3

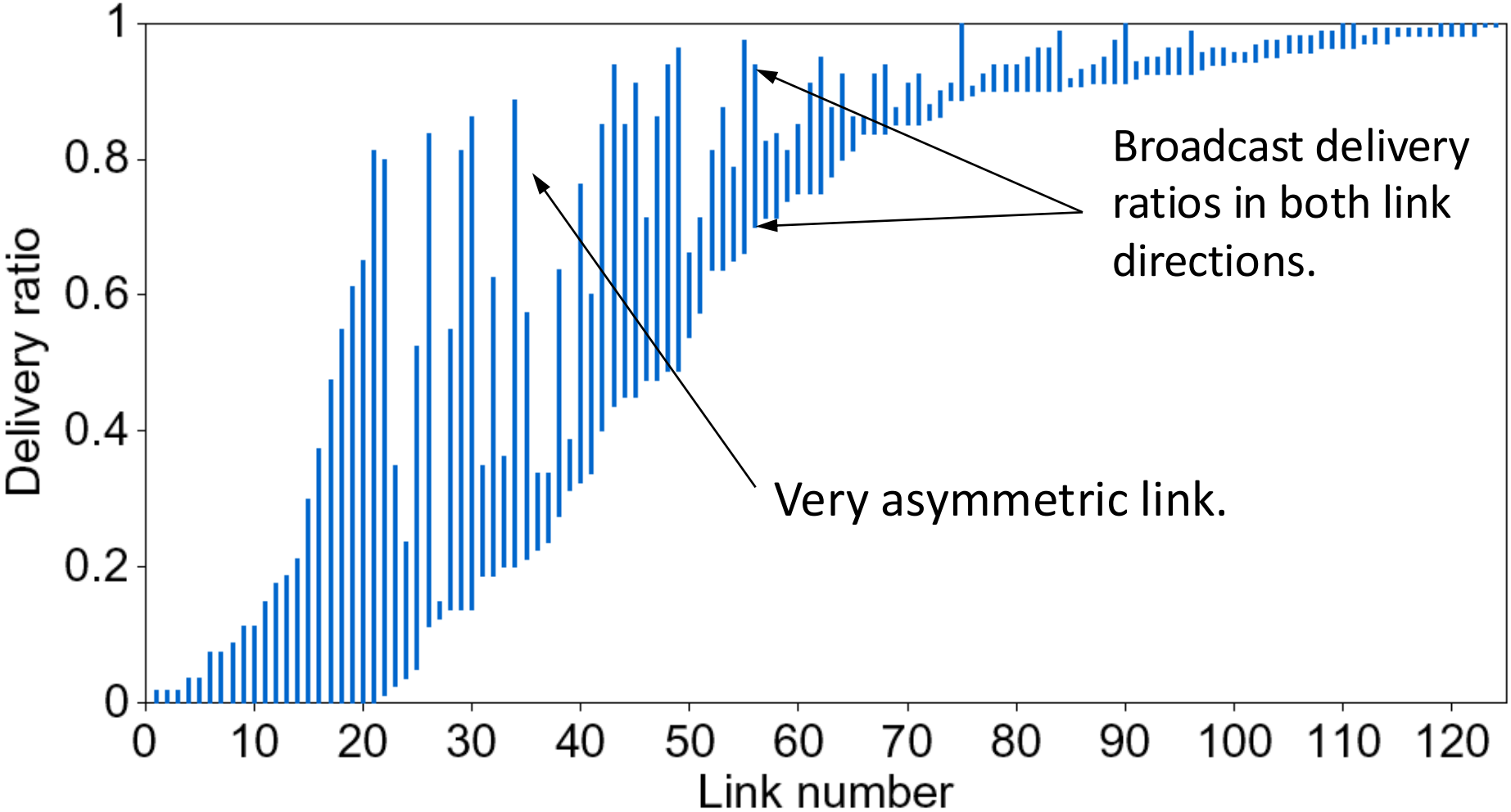
Pros? Cons?

Challenge: many links are lossy



Smooth link distribution complicates link classification.

Challenge: links are lossy and asymmetric

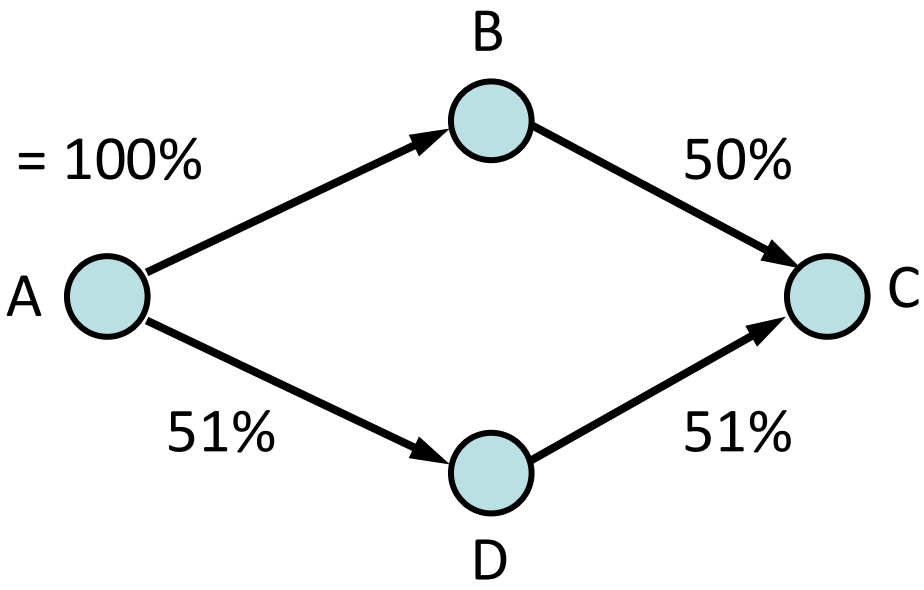


Different links have different loss rates

Further, the loss rate may be different in each direction

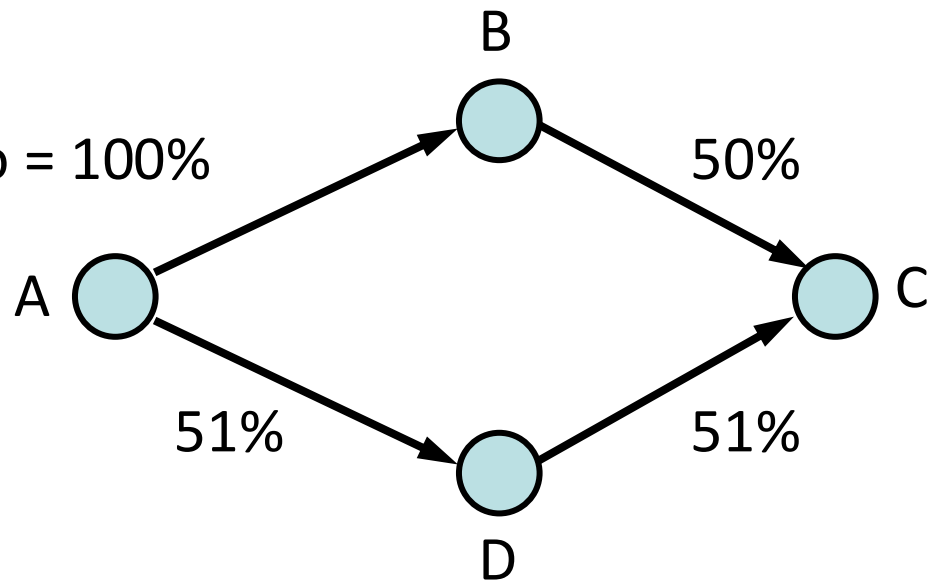
Approach 2:
Maximize bottleneck throughput

Delivery ratio = 100%



Approach 2:
Maximize bottleneck throughput

Delivery ratio = 100%



Bottleneck throughput:

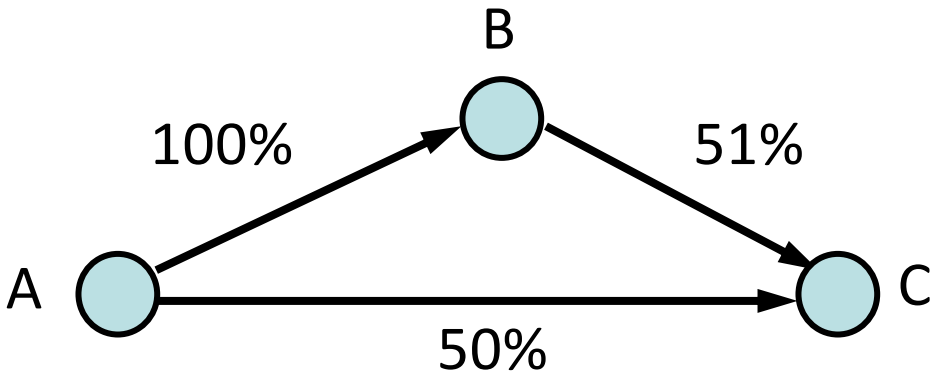
- A-B-C = 50%
- A-D-C = <u>51%</u>

Actual throughput:

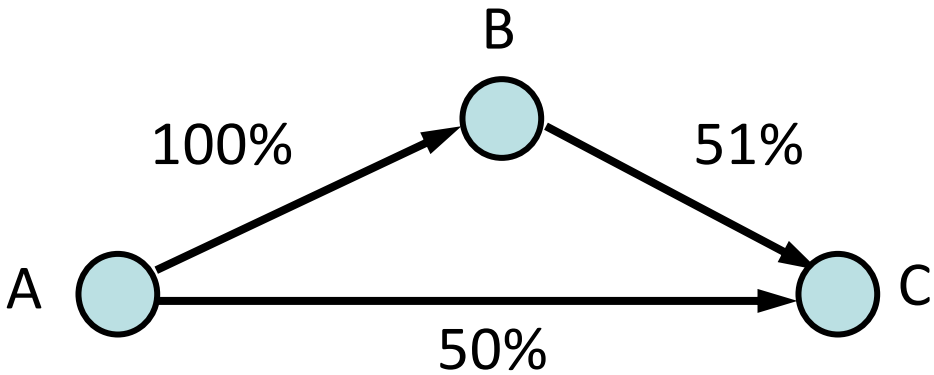
- A-B-C : A✸BA✸BA✸B = <u>33%</u>
- A-D-C : ✸A✸DA✸A✸D = 25%

Pros? Cons?

Approach #3:
Maximize end-to-end delivery ratio



Approach #3:
Maximize end-to-end delivery ratio



End-to-end delivery ratio:

A-B-C = 51%
A-C = 50%

Actual throughput:

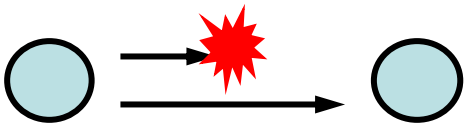
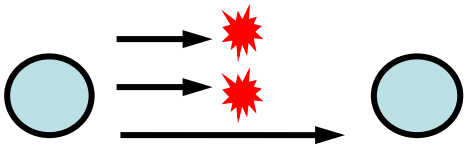
A-B-C : A✸BA✸BA✸B = 33%
A-C : ✸A✸A✸A✸A = 50%

Pros? Cons?

Approach #4: Wireless routing metric: ETX

Minimize total transmissions per packet
(ETX, 'Expected Transmission Count')

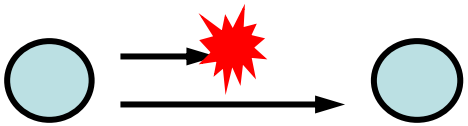
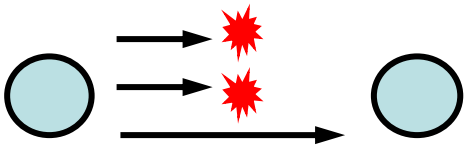
Link throughput ≈ 1/ Link ETX

Delivery Ratio		Link ETX	Throughput
100%			
50%			
33%			

Approach #4: Wireless routing metric: ETX

Minimize total transmissions per packet
(ETX, 'Expected Transmission Count')

Link throughput ≈ 1/ Link ETX

Delivery Ratio		Link ETX	Throughput
100%		1	100%
50%		2	50%
33%		3	33%

Calculating Link ETX

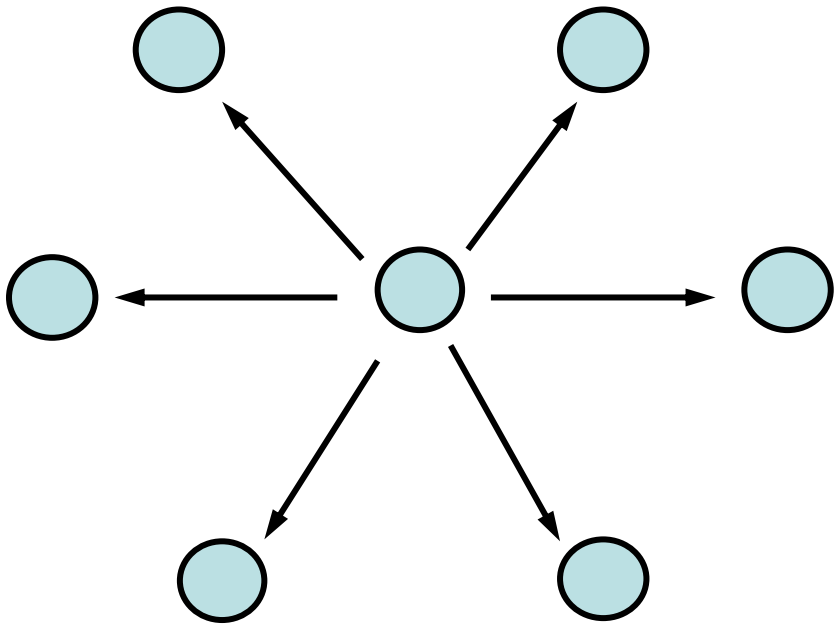
- Assuming 802.11 link-layer acknowledgments (ACKs) and retransmissions:
- P(TX success) = P(Data success) × P(ACK success)
- Link ETX = 1 / P(TX success)
  = 1 / [P(Data success) × P(ACK success)]
- Estimating link ETX:
- P(Data success) ≈ measured fwd delivery ratio r_{fwd}
- P(ACK success) ≈ measured rev delivery ratio r_{rev}
- Link ETX ≈ $1 / (r_{fwd} \times r_{rev})$

How can we measure delivery ratios?

- Each node broadcasts small link probes once per second
- Nodes remember probes received over past 10 seconds
- Reverse delivery ratios estimated as

$$r_{rev} \approx \text{pkts received / pkts sent}$$

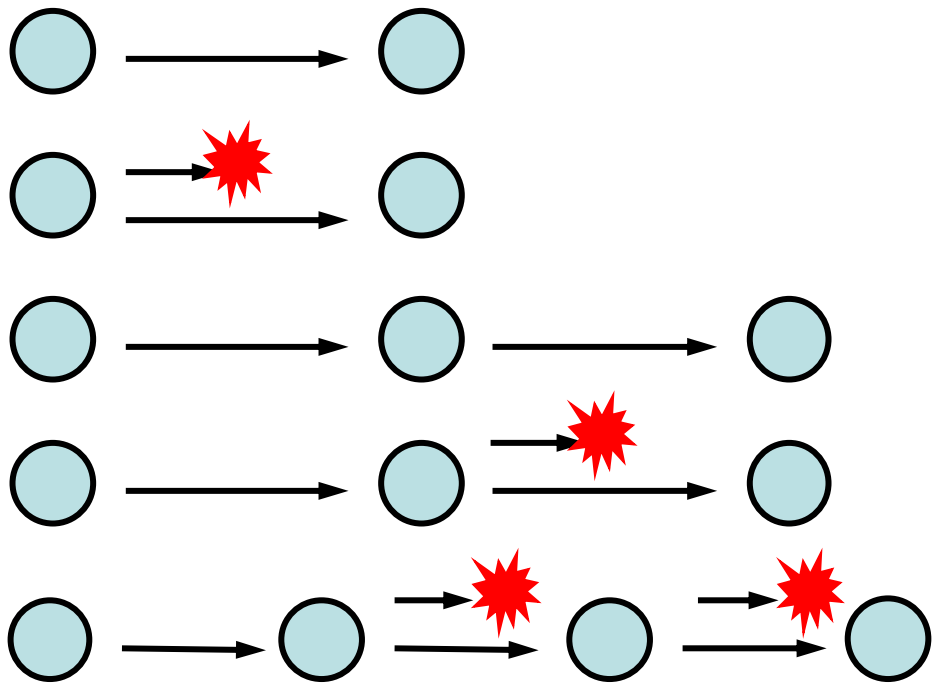
- Forward delivery ratios obtained from neighbors (piggybacked on probes)



Route ETX

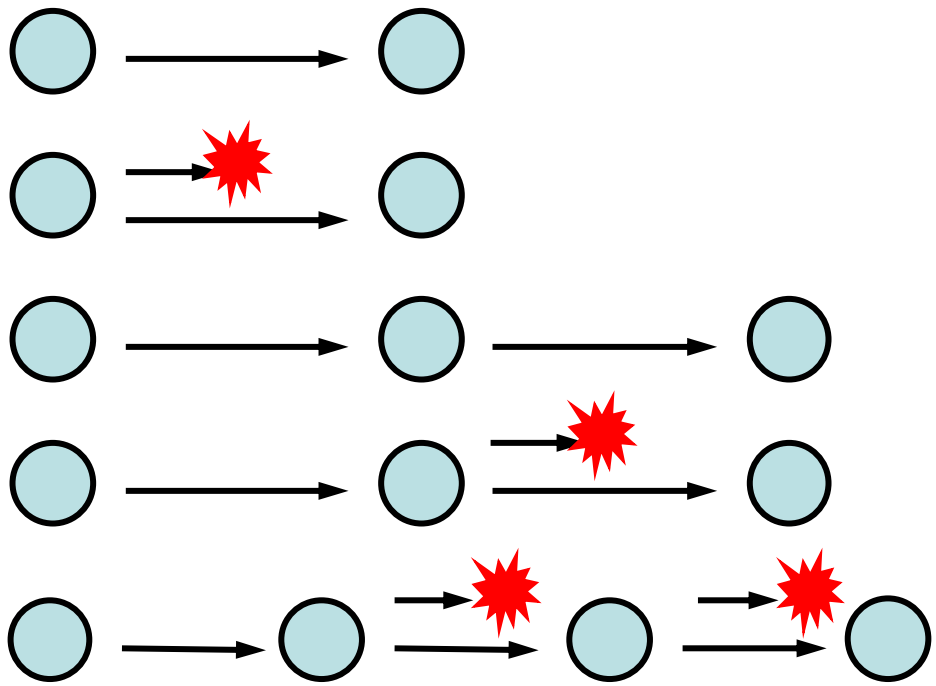
Route ETX = Sum of link ETXs

Route ETX	Throughput



Route ETX

Route ETX = Sum of link ETXs



Route ETX	Throughput
1	100%
2	50%
2	50%
3	33%
5	20%

ETX Pros?

ETX Pros?

- ETX predicts throughput for short routes (1, 2, and 3 hops)
- ETX captures loss
- ETX captures asymmetry

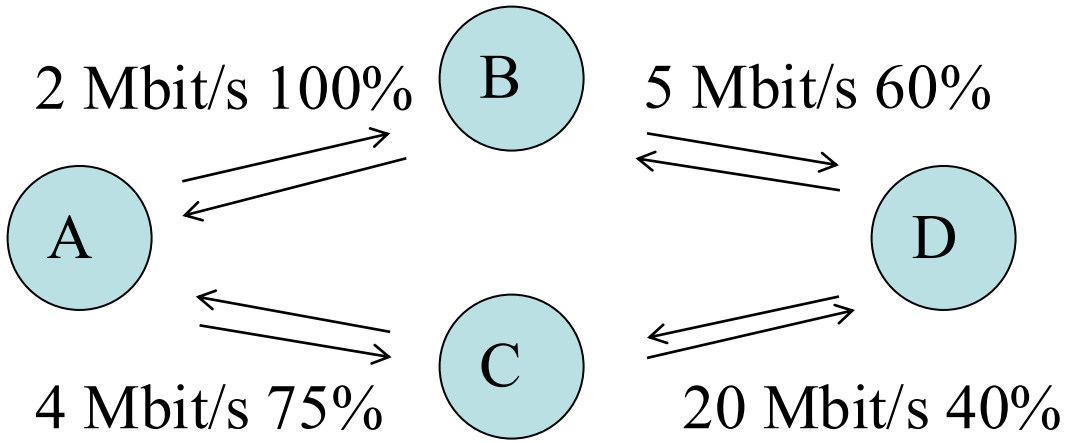
ETX Caveats

ETX Caveats

- It is hard to measure link quality/loss
  - Changes as a function of load
  - Changes with time
- ETX ignores differences in bit-rate and packet size
- ETX ignores spatial re-use (i.e., assumes all links interfere)

How Can We Account to Different Bitrates?

(and different delivery ratios)



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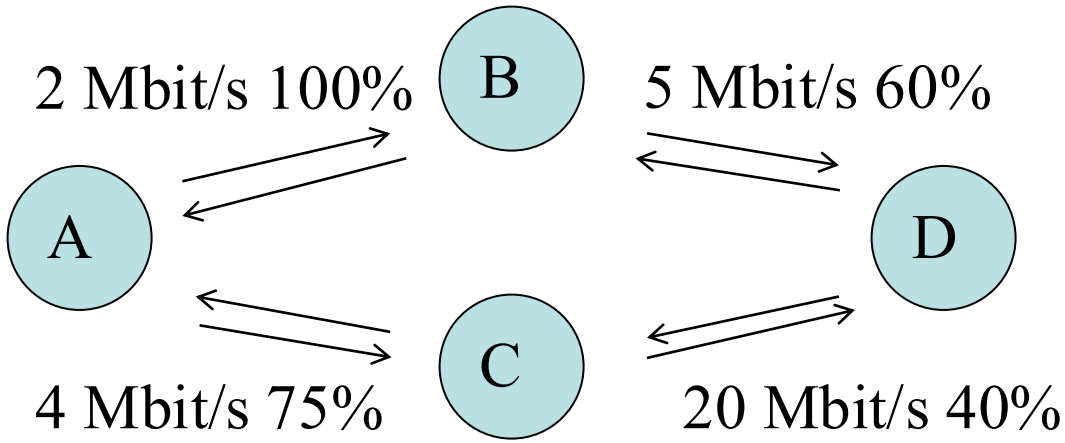


<u>Idea:</u> Take into account both the delivery rate and the **time** taken to transmit packet (i.e., time occupied on "air" by packet)

Assume pkt size = 20

ETT = ETX *(pkt_size/link-bit-rate)

Focus on 1-way

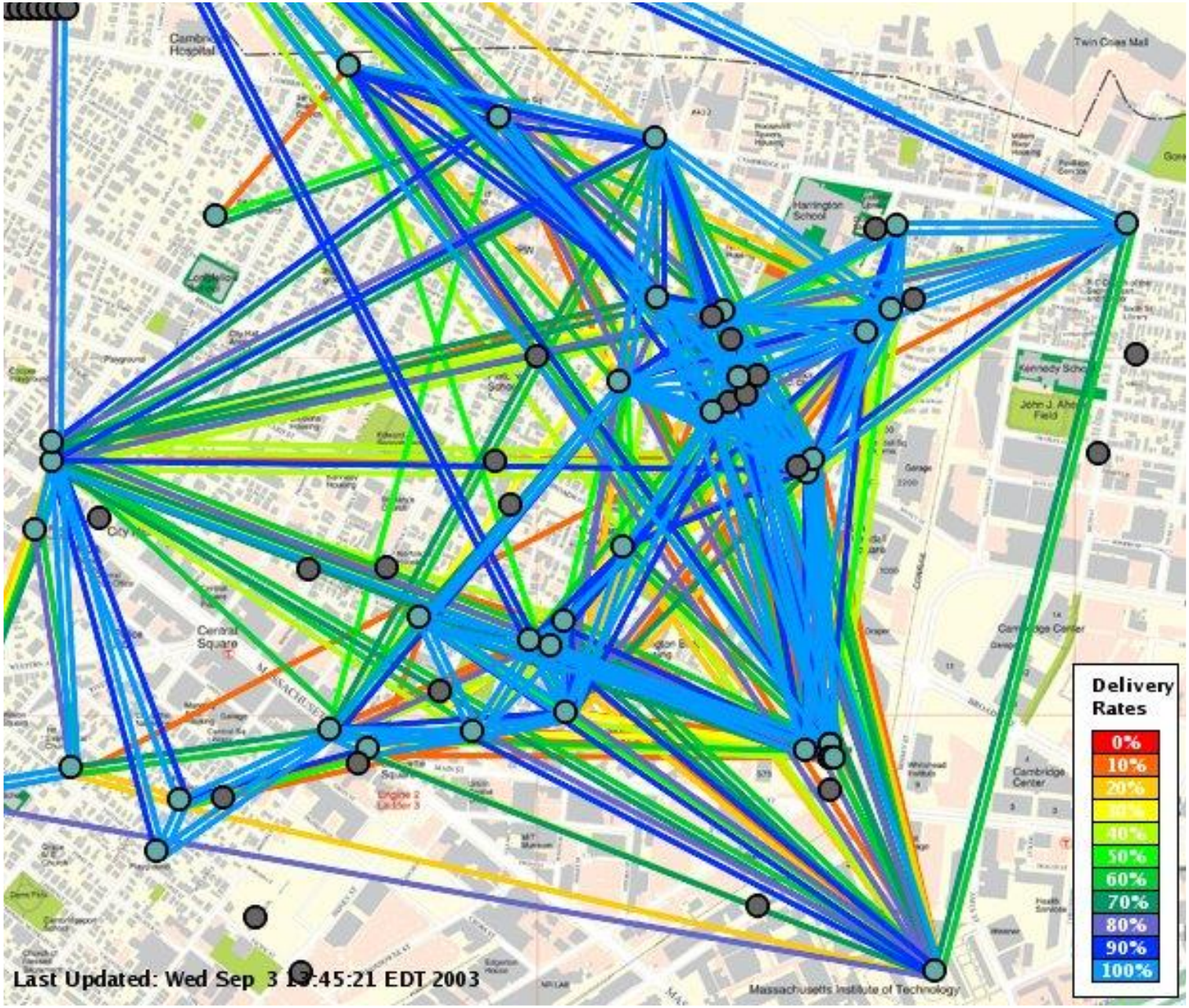
ABD: 1*10+ 5/3*4 = 50/3

ACD: 4/3*5 + 2/5*1 = 55/6

Caveats?

- Bitrate and delivery ratio are related
  - If Tx at higher rate, bitrate is lower
  - In fact, this problem led to the original rise of "information theory"
  - CS/EE -> rate adaptation based on "SNR": signal-to-noise ratio
- Use multiple channels at the same time: each at different bitrate

MIT Roofnet


Delivery Rates
0%
10%
20%
30%
40%
50%
60%
70%
80%
90%
100%
Last Updated: Wed Sep 3 13:45:21 EDT 2003
Massachusetts Institute of Technology
Central Square
Cambridge Center
Twin Cities Mall

Where are mesh networks used today?

- Defense
- Electric car meters
- Home networks (e.g., Google WiFi, TP-link, etc.)
- Satellite constellations

Objectives of the Three Lectures Series

Learn the fundamentals, applications, and implications of

IoT connectivity technologies

1. What is the overall IoT system architecture?
2. What are the various classes of connectivity technologies? And how do we choose the "right" technology for a given application?
3. What are various routing architectures for wireless networks & IoT systems?
4. How does energy impact IoT device design? And how do batteryless IoT systems work?

next lecture